

Cerebellar Hypoplasia: Differential Diagnosis and Diagnostic Approach

ANDREA PORETTI*, EUGEN BOLTSHAUSER, AND DAN DOHERTY

Cerebellar hypoplasia (CH) refers to a cerebellum with a reduced volume, and is a common, but non-specific neuroimaging finding. The etiological spectrum of CH is wide and includes both primary (malformative) and secondary (disruptive) conditions. Primary conditions include chromosomal aberrations (e.g., trisomy 13 and 18), metabolic disorders (e.g., molybdenum cofactor deficiency, Smith–Lemli–Opitz syndrome, and adenylosuccinase deficiency), genetic syndromes (e.g., Ritscher–Schinzel, Joubert, and CHARGE syndromes), and brain malformations (primary posterior fossa malformations e.g., Dandy–Walker malformation, pontine tegmental cap dysplasia and rhombencephalosynapsis, or global brain malformations such as tubulinopathies and α -dystroglycanopathies). Secondary (disruptive) conditions include prenatal infections (e.g., cytomegalovirus), exposure to teratogens, and extreme prematurity. The distinction between malformations and disruptions is important for pathogenesis and genetic counseling. Neuroimaging provides key information to categorize CH based on the pattern of involvement: unilateral CH, CH with mainly vermis involvement, global CH with involvement of both vermis and hemispheres, and pontocerebellar hypoplasia. The category of CH, associated neuroimaging findings and clinical features may suggest a specific disorder or help plan further investigations and interpret their results. Over the past decade, advances in neuroimaging and genetic testing have greatly improved clinical diagnosis, diagnostic testing, recurrence risk counseling, and information about prognosis for patients and their families. In the next decade, these advances will be translated into deeper understanding of these disorders and more specific treatments. © 2014 Wiley Periodicals, Inc.

KEY WORDS: cerebellum; hypoplasia; genetics; neuroimaging; malformations; disruptions

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INTRODUCTION

The term “cerebellar hypoplasia” (CH) is purely descriptive and refers to a cerebellum with a reduced volume, but a normal shape [Boltshauser, 2004; Poretti and Boltshauser, 2012]. In vivo, the diagnosis of CH is based on neuroimaging.

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disorders, genetic syndromes, and brain malformations. These include primary (malformative, genetic) and secondary (disruptive, acquired) lesions [Hennekam et al., 2013]. The distinction between malformations and disruptions is important for pathogenesis and genetic counseling.

The cerebellar involvement is heterogeneous: hypoplasia may affect the entire cerebellum (most commonly) or selectively involve the vermis alone or one/both hemispheres sparing the vermis.

Andrea Poretti received his MD degree in Zurich, Switzerland, where he did his residency in Pediatrics and Pediatric Neurology. He is currently pursuing a postdoctoral research fellowship in pediatric neuroimaging at The Johns Hopkins University in Baltimore, MD.

Eugen Boltshauser earned his MD degree in Zurich, Switzerland, where he was the head of Pediatric Neurology at the University Children's Hospital between 1986 and 2011, as well as a full Professor of Pediatric Neurology.

Dan Doherty earned his MD and PhD degrees at the University of California San Francisco, studying nervous system development in *Drosophila*. Since that time, he has been at the University of Washington, where he is now Associate Professor of Pediatrics in the Divisions of Developmental and Genetic Medicine.

*Correspondence to: Andrea Poretti, Section of Pediatric Neuroradiology, Division of Pediatric Radiology, The Russell H. Morgan Department of Radiology and Radiological Science, The Johns Hopkins School of Medicine, Charlotte R. Bloomberg Children's Center, Sheikh Zayed Tower, Room 4174, 1800 Orleans Street, Baltimore, MD 21287-0842. E-mail: aporetti1@jhmi.edu

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In this article, we (i) discuss general aspects about clinical presentation and neuroimaging findings of CH, (ii) propose a classification of CH based on neuroimaging patterns, and (iii) suggest a diagnostic work-up for CH.

CLINICAL SPECTRUM ASSOCIATED WITH CEREBELLAR HYPOPLASIA

The clinical phenotype associated with CH is wide and depends also on associated brain malformations or additional unrelated symptoms.

Generally, children present with muscular hypotonia and global developmental delay and develop cerebellar signs later [Bolduc and Limperopoulos, 2009]. Neurological findings include truncal ataxia (49–93%), hypotonia (47–49%), ocular movement disorders (40–46%), dysarthria (38%), intention tremor (9–35%), and microcephaly (20%) [Wassmer et al., 2003]. Seizures are more prevalent than in the general population (28–56%) [Ventura et al., 2006].

Intellectual disability is present in more than 60% of patients and is severe in 35% of them [Wassmer et al., 2003]. Speech and language disorders are common and range from mild impairment to total absence of language development. Behavioral abnormalities are also common and 5–20% of patients have autistic features [Wassmer et al.,

2003]. The spectrum of cognitive and behavioral abnormalities matches the cerebellar cognitive affective syndrome [Schmahmann and Sherman, 1998]. This syndrome delineates the contribution of the cerebellum to non-motor functions and includes disturbance of executive function, visuospatial disorganization and impaired visuospatial memory, personality changes and language difficulties.

NEUROIMAGING EVALUATION OF CEREBELLAR HYPOPLASIA

Neuroimaging is mandatory for the diagnosis of CH. Magnetic resonance imaging (MRI) is the neuroimaging tool of choice. Evaluation of global cerebellar and posterior fossa morphology in various imaging planes is essential to the evaluation (Table I) [Poretti et al., 2008c; Doherty et al., 2013]. The size of the cerebellar vermis can be best assessed on midsagittal images. The normal rostrocaudal size of the vermis ranges from the bottom of the inferior colliculus to the obex in a neonate and from the intercollicular sulcus located in the middle between the superior and inferior colliculi of the quadrigeminal plate to the obex in older children; if smaller, the vermis is hypoplastic. CH is often associated with hypoplasia of the pons. This is most likely caused by a reduction in size of the efferent cerebellar pathways

that course through the pons. A small pons in the setting of CH only means that CH is a result of a prenatal event, but does not automatically result in the diagnosis of pontocerebellar hypoplasia as defined by Peter Barth (see below).

Advanced neuroimaging techniques may be useful in selected cases. Diffusion tensor imaging (DTI) provides information about the micro-architecture of cerebellar structures, the course of cerebellar white matter tracts and their connections with other brain structures [Poretti et al., 2013]. Susceptibility, weighted imaging (SWI) is highly sensitive for blood products and calcifications and is helpful in disruptive lesions [Bosemani et al., 2013].

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TABLE I. Role of the MRI Planes for the Evaluation of Cerebellar Hypoplasia

MRI planes	Key points
Coronal	1. Excellent overview of vermis and hemispheres (severity of vermis vs. hemispheric involvement). 2. Normally, cerebellar fissures radiate from the surface towards the cerebellar nuclei on coronal images. 3. The orientation and curvature of the cerebellar fissures is abnormal in cerebellar dysplasia.
Sagittal	1. Assess vermis, brainstem and fourth ventricle morphology, posterior fossa size and supratentorial midline structures. 2. The posterior margin of the brainstem should be almost straight. 3. The fastigium of the fourth ventricle should be just below the midpoint of the ventral pons. 4. The rostrocaudal height of the vermis should be almost equal to the distance from the tectum to the obex. 5. In cerebellar hypoplasia, the fastigium is upwardly displaced and the rostrocaudal height of the vermis is reduced. 6. Major vermis fissures are less prominent or absent in rhombencephalosynapsis.
Axial	1. Assess the morphology of vermis and hemispheres. 2. Essential to assess the supratentorial structures. 3. Normally, the cerebellar fissures have an “onion-like” orientation parallel to the calvarium.

atrophy (CA). CA implies loss of cerebellar parenchyma with secondary enlargement of the interfolial spaces [Poretti et al., 2008c]. In theory, distinction between CH and CA is easy. In practice, however, it may be challenging or even impossible based on a single study. In non-progressive cerebellar ataxia enlarged cerebellar interfolial spaces mimicking CA were reported

[Yapici and Eraksoy, 2005; Turkmen et al., 2006]. CH may be superimposed by CA that is, in congenital disorders of glycosylation. In these situations, clinical course (progressive vs. non-progressive) and additional clinical and neuroimaging findings are helpful in suggesting a specific disease or to plan further investigations and interpret their results.

CLASSIFICATION OF CEREBELLAR HYPOPLASIA

In this article, we use a simple classification based on the neuroimaging pattern. We classify diseases with CH into four groups: (i) unilateral cerebellar hypoplasia, (ii) CH with mainly vermis involvement, (iii) global CH with involvement of both vermis and hemispheres and (iv) pontocerebellar hypoplasia (Table II).

TABLE II. Differential Diagnosis of Cerebellar Hypoplasia Based on the Neuroimaging Pattern

Neuroimaging pattern	Group of diseases	Diseases/anomalies
UCH		PHACE(S) syndrome; familial porencephaly (<i>COL4A1</i> mutation)
CH with mainly vermis involvement	Posterior fossa malformations	Dandy–Walker malformation; Joubert syndrome; rhombencephalosynapsis
	Genetic syndromes	Congenital ocular motor apraxia type Cogan Acrocallosal syndrome; Gillespie syndrome; Beckwith–Wiedemann syndrome; autism-associated chromosome 22q13 terminal deletion
Global CH	Prenatal infections	Congenital cytomegalovirus infection
	Prenatal teratogens	Antiepileptic drugs (phenytoin, valproic acid); retinoic acid; alcohol; cocaine
	Chromosomal anomalies	Trisomy (13, 18, 21); partial trisomy 12q and monosomy 21q; trisomy 17 mosaicism; monosomy 1p36; ring chromosome 6; de novo X;8 translocation; 13q12.3–q14.11 deletion
	Metabolic disorders	Adenylosuccinase deficiency; Smith–Lemli–Opitz syndrome; molybdenum cofactor deficiency and isolated sulfite oxidase deficiency; copper metabolism disease (<i>SLC33A1</i> mutation); Zellweger syndrome; nonketotic hyperglycinemia; mitochondrial disorders (Leigh disease, pyruvate dehydrogenase deficiency); Mucopolysaccharidoses (types I and II)
	Genetic syndromes	Ritscher–Schinzel (3C) syndrome; Hoyeraal–Hreidarsson syndrome; CHARGE syndrome; Endosteal sclerosis; oculocerebrocutaneous (Delleman) syndrome; microcephaly with simplified gyral pattern, epilepsy and permanent neonatal diabetes syndrome (<i>IER3IP1</i> mutation); neurofibromatosis type 1; pseudo-TORCH syndrome; velocardiofacial syndrome; oculodentodigital syndrome; Cohen syndrome; Cri du chat syndrome; Pallister–Killian syndrome; Galloway–Mowat syndrome; Sengers syndrome; <i>OPHN1</i> -related X-linked intellectual disability
PCH	Non-progressive cerebellar ataxias (see Table III)	
	PCH as defined by Barth	PCH types 1–8 (see Rudnik–Schöneborn and Zerres in this issue)
	Cortical malformations	Lissencephaly (<i>RELN</i> , <i>VLDRL</i> , tubulin genes $\gg$ <i>LIS1</i> , <i>DCX</i> , <i>ARX</i>); polymicrogyria (tubulin genes, <i>GPR56</i>); periventricular nodular heterotopia (<i>FLNA</i>); primary microcephaly
	Metabolic diseases	Congenital disorders of glycosylation (mostly type 1a, but also in type Iq)
	Genetic disorders	<i>CASK</i> mutation; cerebellar agenesis (<i>PTF1A</i>)
	α -dystroglycanopathies	Walker–Warburg syndrome; muscle–eye–brain disease; Fukuyama disease
	Posterior fossa malformations	Pontine tegmental cap dysplasia
	Disruptive lesions	Cerebellar agenesis; cerebellar injury secondary to prematurity

CH, cerebellar hypoplasia; PCH, pontocerebellar hypoplasia; UCH, unilateral cerebellar hypoplasia.

Diseases included in this article were compiled from our clinical experience with children with cerebellar disorders, searches in PubMed and textbooks on cerebellar disorders, ataxia and neuroimaging. This list, however, does not aim to be complete. For several disorders, cerebellar involvement is reported only in few patients or neuroimaging findings are not shown and/or not accurately described. Additionally, some diseases were reported first only recently and their phenotypic spectrum is still expanding. Therefore, the full neuroimaging spectrum is still unknown for most disorders.

Unilateral Cerebellar Hypoplasia

Unilateral cerebellar hypoplasia (UCH, Fig. 1) is rare and ranges from complete aplasia to mild asymmetry in size of the cerebellar hemispheres [Poretti et al., 2010]. UCH may be an incidental finding or identified in patients with developmental delays, intellectual disability, and cerebellar signs.

In the hypoplastic hemisphere, abnormal foliation and/or clefts may be present and associated supratentorial destructive lesions such as schizencephaly have been reported in some patients [Poretti et al., 2008b]. UCH

can develop after a second trimester or early third trimester prenatal cerebellar hemorrhage [Malingier et al., 2006, 2009]. Generally, UCH is acquired, not genetic in origin and represents a residual change after a disruptive prenatal cerebellar insult. This is important for genetic counseling, with a low recurrence risk likely. A genetic predisposition to disruption such as mutations in *COL4A1* may be present [Vermeulen et al., 2011].

Posterior fossa anomalies are present in 40–50% of patients with PHACE(S) syndrome (OMMIM 606519) [Oza et al., 2008; Hess et al., 2010]. UCH with or without involvement of the cerebellar vermis is present in about 75% of PHACE(S) patients with posterior fossa involvement, while a Dandy–Walker malformation is present in less than 5% of patients. In PHACE(S), UCH is almost always associated with abnormalities of the ipsilateral internal carotid artery or a persistent embryonic carotid-basilar connection [Hess et al., 2010]. This association suggests a disruptive origin.

UCH has been reported in two patients with osteogenesis imperfecta and mutations in *WNT1* [Laine et al., 2013; Pyott et al., 2013]. Although *WNT1* plays a role in the cerebellar

development, a causal relation between *WNT1* mutations and UCH needs to be proven.

CH With Mainly Vermis Involvement

Selected diseases, mostly malformations, are characterized by CH with mainly vermis involvement (Fig. 2). In these diseases, cerebellar hemispheres may be normal, reduced in size or even enlarged as in Joubert syndrome [Poretti et al., 2011].

Dandy–Walker malformation

Dandy–Walker malformation (DWM, OMIM 220200) is defined by hypoplasia (or agenesis) of the cerebellar vermis, which is elevated and rotated anticlockwise, and cystic dilatation of the fourth ventricle which extends posteriorly filling out nearly the entire posterior fossa [Parisi and Dobyns, 2003]. Elevation of tentorium and torcula and hydrocephalus may be present. The cerebellar hemispheres are displaced rostrally and laterally, but their volume is often reduced. The brainstem may be hypoplastic. In 30–50% of patients, DWM is associated with additional malformations (e.g., callosal dysgenesis, encephaloceles, and migrational abnormalities) [Parisi and Dobyns, 2003]. Abnormal vermian lobulation and additional brain malformations are unfavorable prognostic factors for cognitive outcome [Boddaert et al., 2003].

A detailed neuroimaging assessment makes it possible to distinguish DWM from Blake's pouch cyst (BPC), arachnoid cysts and mega cisterna magna (MCM) [Poretti et al., 2012a,b]. In BPC and MCM, the cerebellum is normal. The normal size of the fourth ventricle differentiates MCM from DWM and BPC. Mass effect on the cerebellum and obstruction of the fourth ventricle support the diagnosis of arachnoid cysts. Neuroimaging diagnostic criteria enable accurate diagnoses and to avoid terms such as “Dandy–Walker variant,” “Dandy–Walker complex” or “Dandy–Walker spectrum.” Historically, a variety of definitions have been used for these categories, leading to a lack of specificity and confusion over the “true” DWM.

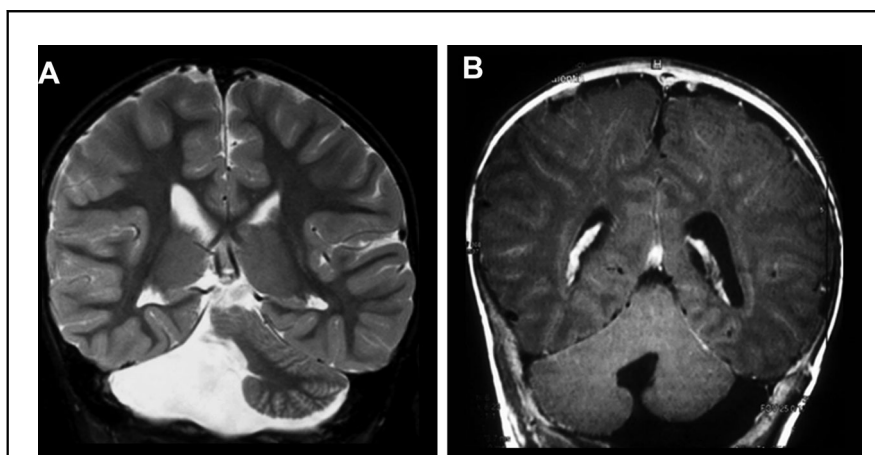


Figure 1. Unilateral cerebellar hypoplasia. **A:** Coronal T2-weighted image shows a total aplasia of the right cerebellar hemisphere in a 4-year-old child found in investigation of developmental delay (reprinted with permission from Poretti et al., 2009b). **B:** Coronal T1-weighted image show left unilateral cerebellar hypoplasia in a child with PHACE(S) syndrome (reprinted with permission from Poretti and Boltshauser, 2012).

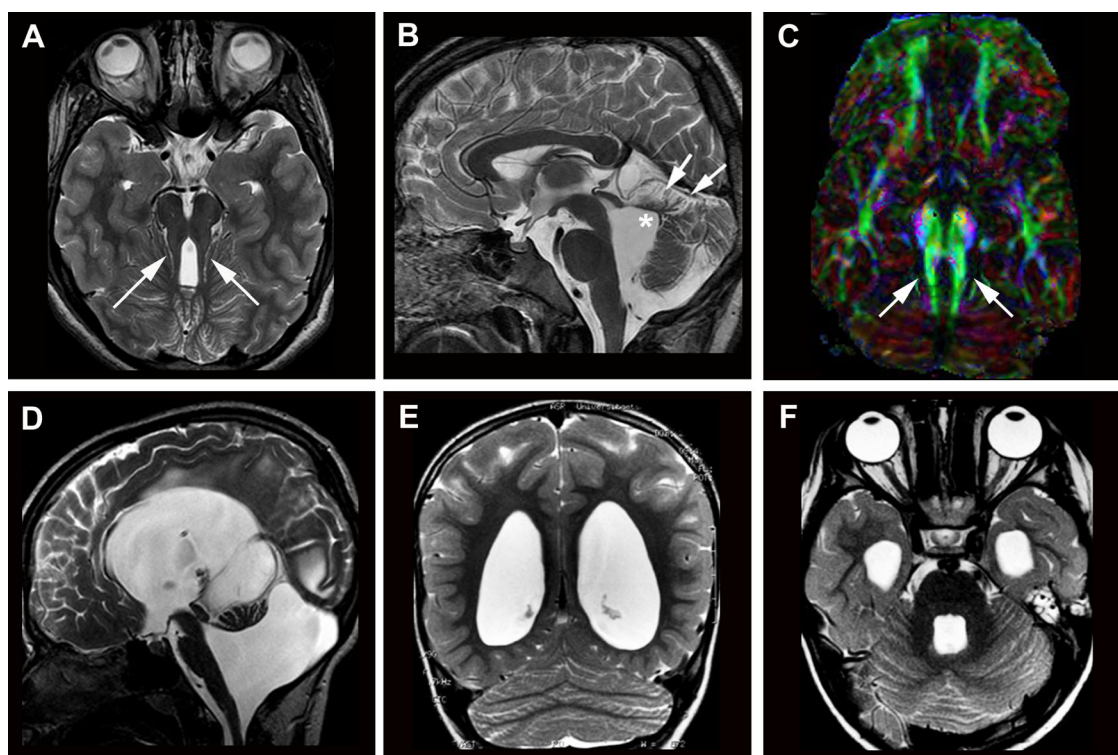


Figure 2. Cerebellar hypoplasia with mainly vermis involvement. **A:** Axial T2-weighted image in a child with Joubert syndrome shows the molar tooth sign with elongated, thickened, and horizontally orientated superior cerebellar peduncles (arrows) and a deepened interpeduncular fossa. **B:** Midsagittal T2-weighted image of the same patient demonstrates severe hypoplasia of the cerebellar vermis with dysplasia of the vermian remnants (arrows), shortening of the ponto-mesencephalic isthmus and enlargement of the fourth ventricle with upwards displacement of the fastigium (asterisk). **C:** Axial color-coded fractional anisotropy maps (DTI) of the same patient reveals the superior cerebellar peduncles (arrows) as encoded in green (green = anterior ↔ posterior orientation), while physiologically the superior cerebellar peduncles are encoded in blue (blue = superior ↔ inferior orientation); additionally, the midline “red dot” (red = right ↔ left orientation) is missing representing absence of decussation of the superior cerebellar peduncles (reprinted with permission from Poretti et al., 2007). **D:** Midsagittal T2-weighted image of a child with Dandy-Walker malformation shows a hypoplastic, elevated and anticlockwise rotated vermis, cystic dilatation of the fourth ventricle with posterior extension and communication with an enlarged posterior fossa and supratentorial hydrocephalus. **E:** Posterior coronal T2-weighted image of a girl with rhombencephalosynapsis demonstrates fused cerebellar hemispheres with an abnormal, transverse orientation of cerebellar folia and ventriculomegaly (reprinted with permission from Poretti et al., 2009a). **F:** Axial T2-weighted image of the same patient reveals fusion of the cerebellar hemispheres without an intervening vermis and dilatation of the temporal horns of the lateral ventricles (reprinted with permission from Poretti et al., 2009a).

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DWM may occur in isolation, in combination with other malformations, or as part of a Mendelian syndrome (e.g., Ritscher-Schinzel, Fryns, or Ellis-van Creveld syndromes). It can be caused by chromosomal anomalies (e.g., trisomy 9, 13, and 18), partial duplications and deletions (e.g., duplication 3q, deletion 3q25.1-25.33, deletion 13q) [Liao et al., 2012].

DWM occurs mostly sporadically and overall the recurrence risk is low (1–5%) [Murray et al., 1985]. Mutations in six genes (*ZIC1*, *ZIC4*, *FOXC1*, *FGF17*, *LAMC1*, and *NID1*) were found in only a small proportion of patients with DWM [Poretti et al., 2012a; Darbro et al., 2013]. The

function of these genes suggests that DWM results from an abnormal interaction between developing cerebellum and developing posterior fossa mesenchyme.

Joubert syndrome

Joubert syndrome (JS) is characterized by hypotonia, ataxia, ocular motor apraxia, neonatal breathing dysregulation, and intellectual disability [Doherty, 2009; Romani et al., 2013]. Systemic involvement may be present and includes kidneys (nephronophthisis 25%), eyes (retinal dystrophy 30% and colobomas 20%), liver (congenital hepatic fibrosis 15%), and skeleton (different forms of polydactyly 20%) [Doherty, 2009].

Renal and liver involvement may cause high morbidity and mortality and needs appropriate work-up and regular follow-up. Based on the systemic involvement, six phenotypes have been described: (i) pure JS (purely neurological involvement), (ii) JS with eye involvement, (iii) JS with kidney involvement, (iv) JS with involvement of eyes and kidneys, (v) JS with liver involvement, and (vi) JS with oral-facial-digital involvement or the oral-facial-digital syndrome type VI (OFDVI) [Romani et al., 2013]. The presence of tongue hamartoma, additional frenula, upper lip notch, mesoaxial polydactyly of one or more hands or feet and/or hypothalamic hamartoma differentiate OFDVI from the other phenotypes [Poretti et al., 2012c]. This clinical-genetic classification is preferable and less confusing (particularly for families) compared to a distinction between JS, JS and related disorders (JSRD) and the individual syndromes included in JSRD (e.g., Dekaban-Arima, Malta, and other syndromes).

Neuroimaging showing the molar tooth sign (MTS) is pathognomonic for the diagnosis of JS. The MTS is characterized by elongated, thickened and horizontally orientated superior cerebellar peduncles (SCP) and a deep interpeduncular fossa. Hypoplasia and dysplasia of the vermis is another consistent finding. The spectrum of neuroimaging findings is beyond the MTS and vermian hypoplasia and dysplasia, confirming the heterogeneity of JS [Poretti et al., 2011]. The cerebellar hemispheres may have variable size and morphology. Morphological abnormalities of the brainstem and supratentorial involvement (e.g., callosal dysgenesis, cephaloceles, and migrational disorders) occur in about 30% of the patients. Neuroimaging is of limited value in classifying JS patients. Differences in neuroimaging findings were reported in siblings and only a weak neuroimaging-genotype correlation was found [Poretti et al., 2011]. Only the presence of a hypothalamic hamartoma differentiates between OFDVI and the other phenotypes [Poretti et al., 2012c]. DTI can demonstrate the absence of SCP and

corticospinal tract decussation, which implies an underlying axonal guidance defect [Poretti et al., 2007].

Twenty-four genes have been associated with JS so far [Halbritter et al., 2013; Romani et al., 2013; Thomas et al., 2014; Tuz et al., 2014]. Some degree of genotype-phenotype correlation has been shown. The strongest correlation is between mutations in *TMEM67* and liver involvement [Doherty et al., 2010]. Mutations in all the genes but *OFD1* are autosomal recessively inherited. This results in a recurrence risk of 25%. All genes encode for proteins that localize to primary, non-motile cilia, and its basal body which play key roles in the development and functioning of the brain, retina, kidney, liver, and other organs [Hildebrandt et al., 2011]. Primary cilia mediate various signaling processes and brain malformations in JS may result from defects in midline fusion of the developing vermis or defects in sonic hedgehog-mediated granule cell proliferation [Lancaster et al., 2011; Aguilar et al., 2012].

Rhomboencephalosynapsis

Rhomboencephalosynapsis (RES) is characterized by absence of the vermis and fusion of the cerebellar hemispheres and is thought to be caused by aberrant

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dorsal-ventral patterning [Ishak et al., 2012]. The posterior coronal sections are

crucial to evaluate the horizontal cerebellar folial pattern. The entire cerebellar volume is reduced only in the most severe cases. RES is often associated with midbrain abnormalities (e.g., midline fusion of the colliculi and aqueductal stenosis), hydrocephalus, and supratentorial abnormalities (absent septum pellucidum, callosal dysgenesis, and holoprosencephaly).

Children present with truncal and/or limb ataxia, abnormal eye movements and delayed motor development. About 85% of patients have head-shaking stereotypies (repetitive figure-8 or side-to-side swinging motion) that may represent a response to deficits in central vestibular processing [Tully et al., 2013]. Long-term cognitive outcome varies between severe impairment to normal and seems to correlate with the severity of neuroimaging findings [Poretti et al., 2009a; Ishak et al., 2012].

The majority of patients with RES do not have other syndromic findings. RES is, however, a key feature of Gómez-López-Hernández syndrome (OMIM 601853, parietal alopecia, trigeminal anesthesia, and craniofacial dysmorphic signs) [Poretti et al., 2008a; Sukhudyay et al., 2010] and features of VACTERL association (Vertebral anomalies, Anal atresia, Cardiovascular anomalies, Trachea-Esophageal fistula, Renal anomalies, Limb defects) are seen in some patients with RES [Tully et al., 2012]. RES is a sporadic malformation with no familial recurrence.

Congenital ocular motor apraxia type Cogan

Congenital ocular motor apraxia type Cogan is characterized by the inability to initiate volitional horizontal saccades [Salman and Ikeda, 2013]. Children present during early infancy with attenuated visual response and compensatory rapid thrusting head movements. Truncal ataxia and intellectual disability may be present. Neuroimaging may be normal or show CH and other abnormalities (e.g., callosal agenesis) [Sargent et al., 1997]. There is no correlation between neuroimaging findings and clinical outcome.

Global CH With Involvement of Both Vermis and Hemispheres

Global cerebellar hypoplasia refers to involvement of both vermis and cerebellar hemispheres. This is the largest group of diseases associated with CH (Fig. 3). We will categorize them according to their etiology.

Congenital infections

Global CH was reported in children with congenital infections, particularly cytomegalovirus (CMV) [Poretti et al., 2009b]. The spectrum of neuroimaging findings includes multifocal white matter lesions that affect predominantly the temporo-parietal regions, migrational abnormalities, ventriculomegaly, calcifications, and subcortical cysts. Migrational disorders and cerebellar involvement are more severe with infection earlier in pregnancy. Lissencephaly and severe CH are found in early infections (before 16–18 weeks of gestation), while polymicrogyria and mild CH are seen in later infections (18–24 weeks of gestation). In third trimester infections the cerebellum is normal [Steinlin et al., 1996]. Calcifications are located in periventricular regions, basal ganglia and cerebellar white matter and are best seen on SWI.

The entire spectrum of neuroimaging findings and the clinical presentation (e.g., lethargy, poor feeding, seizures, hyper- or hypotonia, microcephaly, chorioretinitis, and sensorineural deafness) may suggest the diagnosis. Because postnatal CMV infection is common, diagnostic testing should be performed within the first three postnatal weeks. A positive test thereafter is non-conclusive. If the patient presents beyond the neonatal period, the diagnosis may be confirmed by presence of CMV-DNA in dried blood of neonatal screening (Guthrie). This diagnosis is important because of emerging evidence that postnatal antiviral treatment can improve outcomes [Oliver et al., 2009].

Prenatal exposure to teratogens

Global CH was reported after prenatal exposure to anticonvulsant drugs (e.g., phenytoin and valproic acid) [Squier

et al., 1990], retinoic acid, alcohol [Norman et al., 2009], and cocaine [Bellini et al., 2000]. History of the pregnancy and presence of dysmorphic findings (fetal alcohol syndrome) are leading factors to the diagnosis.

Chromosomal abnormalities

Global CH was inconsistently reported in children/fetuses with various chromosomal abnormalities. Trisomies 13 and 18 are the chromosomal abnormalities most consistently associated with CH [Rosati and Guariglia, 1999; Lin et al., 2006]. Less consistent associations include for example, trisomy 21, partial trisomy 12q and monosomy 21q, trisomy 17 mosaicism, monosomy 1p36, ring chromosome 6, X;8 translocation and sex chromosomal anomalies (e.g., X monosomy) [Ulgiate et al., 2013].

Metabolic disorders

CA is the most common cerebellar involvement in neurometabolic disorders [Poretti et al., 2008c]. Global CH, however, has been reported only in a small number of neurometabolic disorders.

Adenylosuccinase deficiency (OMIM 103050) is a disease of purine metabolism. The clinical presentation is variable and includes developmental delay, intellectual disability, autistic features, hypotonia, and seizures. The diagnosis is made by elevated succinyl nucleosides in body fluids and can be genetically confirmed by mutations in *ADSL*. Neuroimaging findings are non-specific and include global CH at an early stage and diffuse white matter signal abnormalities, cerebral atrophy, and CA at a later stage [Edery et al., 2003].

Smith–Lemli–Opitz syndrome (SLO, OMIM 270400) is an autosomal recessively inherited disorder that is caused by mutations in *DHCR7*, resulting in impaired cholesterol synthesis. SLO is a multisystem disease and its clinical presentation is broad. The most common clinical features are cognitive impairment, anteverted nares, postnatal growth retardation, syndactyly of the second and third toe, microcephaly, ambiguous genitalia in males, ptosis, cleft palate and congenital heart defects.

The diagnosis is made by increased 7-dehydrocholesterol in blood and can be genetically confirmed. The most common neuroimaging findings are callosal dysgenesis (hypoplasia, thickening, or abnormal shape), colpocephaly, arachnoid cysts, and white matter lesions [Lee et al., 2013]. CH of variable severity (global or with prominent involvement of the inferior vermis) is common [Caruso et al., 2004].

Molybdenum cofactor deficiency (OMIM 252150) and isolated sulfite oxidase deficiency (OMIM 272300) are autosomal recessive disorders of the catabolism of sulphur-containing aminoacids. The accumulated sulfites are toxic for mitochondria in the brain. Affected patients typically present in the first days of life with feeding difficulties, vomiting, and intractable seizures. The neuroimaging findings show temporal alterations and include initially global cerebral edema and later cystic encephalomalacic changes in the subcortical white matter and signal abnormalities in the basal ganglia [Sass et al., 2010]. Global CH is common [Carmi-Nawi et al., 2011].

Finally, global CH has been occasionally reported in Zellweger syndrome, nonketotic hyperglycinemia, mitochondrial disorders (e.g., Leigh disease and pyruvate dehydrogenase deficiency), and mucopolysaccharidoses.

Genetic syndromes

Global CH has been reported as a typical, occasional or rare finding in multiple genetic syndromes.

Ritscher–Schinzel (3C = cranio-cerebello-cardiac) syndrome (OMIM 220210) is characterized by craniofacial

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is the most common posterior***

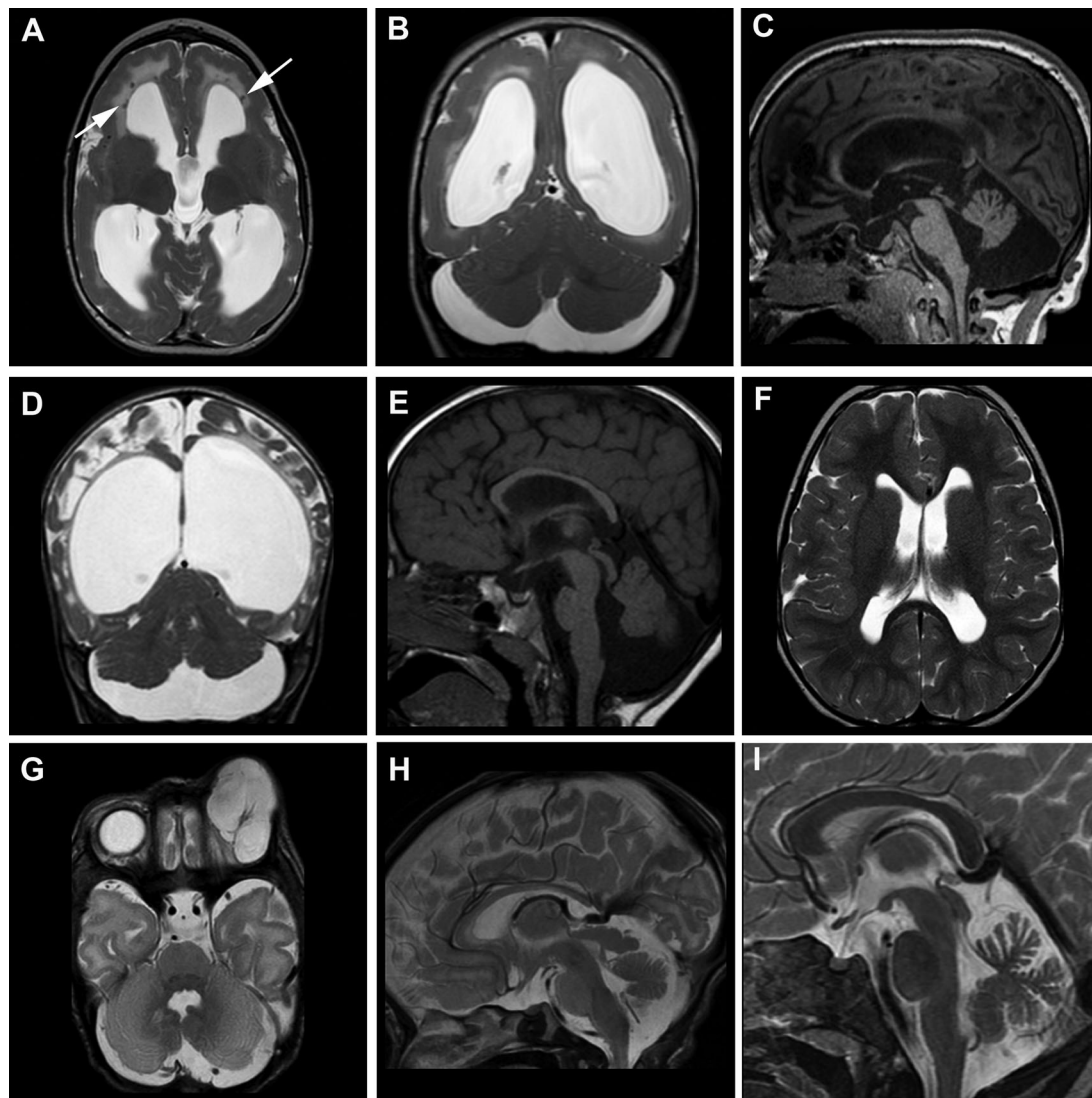


Figure 3. Global cerebellar hypoplasia with involvement of both vermis and hemispheres. **A:** Axial T2-weighted image of a 14-month-old child with confirmed prenatal CMV infection shows marked ventriculomegaly, diffuse hyperintense signal of the supratentorial white matter, generalized cortical malformation including pachygyria and polymicrogyria and periventricular hypointense spots suggestive of calcifications (arrows) (reprinted with permission from Poretti et al., 2009b). **B:** Coronal T2-weighted image of the same child demonstrates global mild cerebellar hypoplasia, marked ventriculomegaly, diffuse hyperintense signal of the supratentorial white matter and generalized cortical malformation including pachygyria and polymicrogyria (reprinted with permission from Poretti et al., 2009b). **C:** Midsagittal T1-weighted image of a 9-year-old child with mucopolysaccharidosis type II reveals cerebellar hypoplasia, enlargement of the fourth ventricle, short midbrain, ventriculomegaly, enlargement of the pituitary sella, and thickening of the diploic space (reprinted with permission from Alqahtani et al., 2014). **D:** Coronal T2-weighted image of a child with confirmed molybdenum cofactor deficiency shows global cerebellar hypoplasia and cerebral atrophy with ulegyric pattern and marked ventriculomegaly. **E:** Midsagittal T1-weighted image of a 22-month-old girl with *TUBA1A* mutation shows cerebellar and pontine hypoplasia with abnormally higher rostrocaudal length of medulla and midbrain with respect to the pons and loss of the normal flat dorsal surface of the brainstem. **F:** Axial T2-weighted image of the same child demonstrates dysmorphic basal ganglia with loss of a clear interface between them and internal capsule and bilateral ventriculomegaly with abnormal configuration of the anterior horns of the lateral ventricles. **G:** Axial T2-weighted image of a 6-day-old neonate with oculocerebrocutaneous syndrome shows a cystic malformation of the left eye and global cerebellar hypoplasia (reprinted with permission from Poretti and Boltshauser, 2012). **H:** Midsagittal T2-weighted image of the same neonate reveals cerebellar hypoplasia and enlargement and dysplasia of the tectum (reprinted with permission from Poretti and Boltshauser, 2012). **I:** Midsagittal T2-weighted image of a teenager with non-progressive cerebellar ataxia and a static course over several years shows dilated interfoliar spaces mimicking cerebellar atrophy.

fossa malformation associated with this syndrome (about 65–70% of the patients), while global CH is reported in 20–25%.

dysmorphisms, congenital heart defects and posterior fossa malformations [Leonardi et al., 2001; Elliott et al., 2013]. DWM is the most common posterior fossa malformation associated with this syndrome (about 65–70% of the patients), while global CH is reported in 20–25% [Leonardi et al., 2001].

Hoyeraal-Hreidarsson syndrome (OMIM 300240) is an X-linked multi-system disorder caused by mutations in *DKC1*. Children present with prenatal onset growth failure, secondary microcephaly, developmental delay, ataxia, immunodeficiency, and progressive bone marrow failure with aplastic anemia [Sznajder et al., 2003]. Global CH is consistently reported. Other neuroimaging findings include delayed myelination and callosal hypoplasia.

CHARGE syndrome (OMIM 214800) is an autosomal dominant disorder characterized by Colobomas, Heart malformations, Atresia of choanae, Retarded mental development, Genital anomalies, and Ear anomalies with or without deafness. Brain abnormalities were reported in 55–85% of the patients [Pinto et al., 2005]. The most common findings are hypoplastic olfactory bulbs and holoprosencephaly. DWM and global CH were occasionally reported. Most cases are caused by de novo mutations in *CHD7*, that disrupt early neural tube development and result in CH [Yu et al., 2013].

Endosteal sclerosis (OMIM 213002) is characterized by increased radiodensity of bones and seems to be consistently associated with CH [Ozgen et al., 2005]. Clinical findings are ataxia, nystagmus, microcephaly, short stature, tooth eruption disturbances, optic atrophy, and congenital hip dislocation.

Oculocerebrocutaneous syndrome or Delleman syndrome (OMIM 164180) is characterized by eye (anoph-

thalmia, microphthalmia, colobomas), brain and skin abnormalities (skin appendages, focal skin aplasia, or hypoplasia). A giant and dysplastic midbrain tectum is a distinct neuroimaging finding [Moog et al., 2007]. Other posterior fossa abnormalities include global CH, cerebellar dysplasia, and large posterior fossa fluid collections. Abnormal definition of the midbrain-rhombomere 1 boundary has been proposed as the mechanism for the tectal and cerebellar abnormalities.

Global CH has also been reported occasionally in neurofibromatosis type 1, pseudo-TORCH syndrome, oculodentodigital syndrome, and Cohen syndrome. Global CH has been reported rarely in Cri du chat, Pallister-Killian, Galloway-Mowat, and Sengers syndromes.

Non-progressive cerebellar ataxias

Non-Progressive Cerebellar Ataxia (NPCA) refers to children with early evidence of cerebellar ataxia without progression on follow-up [Boltshauser and Poretti, 2012]. First features of ataxia are manifested between 2 and 3 years of age, and are preceded by hypotonia and delayed motor and language milestones [Steinlin et al., 1998]. Cognitive impairment is common and is the most limiting factor in older children and young adults [Steinlin et al., 1999].

Neuroimaging is variable and inter- and intra-familial variability is common. In a large proportion of children, the brain MRI is normal. In a subgroup of patients, neuroimaging reveals a small cerebellum with widened interfolial spaces, giving the impression of CA [Yapici and Eraksoy, 2005; Turkmen et al., 2006]. In view of the static clinical course, we consider this cerebellar appearance as a form of CH. This observation implies that a single neuroimaging study may not be able to distinguish a progressive CA from a static CH [Poretti et al., 2008c]. Finally, in a small number of patients neuroimaging shows a classic CH (reduced cerebellar volume with normal architecture).

In NPCA, familial recurrence has been reported repeatedly, and a genetic

basis is assumed (Table III). In recent years, an increasing number of genes have been associated with NPCA, indicating marked genetic heterogeneity. Recessive inheritance is the most common, with X-linked and autosomal dominant forms also described.

Pontocerebellar Hypoplasia

The use of the term pontocerebellar hypoplasia (PCH) is not uniform. It is often used in a non-specific descriptive manner to imply volume reduction of cerebellum and pons. However, it was also used in the context of the so-called pontocerebellar hypoplasias, as previously conceptualized by Peter Barth. This

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heterogeneous group includes prenatal onset degenerative disorders (types 1, 2, 4, 5, 6, and 7) and diseases with a non-progressive course (types 3 and 8) and will be discussed by Rudnik-Schöneborn et al., in another article of this issue. PCH can be observed in several other conditions (Fig. 4).

Malformations of cortical development

Cerebellar involvement including PCH was reported in a wide spectrum of malformations of cortical development including lissencephaly, polymicrogyria, periventricular nodular heterotopia (*FLNA* mutations) [Pisano et al., 2012], and primary microcephaly [Adachi et al., 2014].

TABLE III. Non-progressive Cerebellar Ataxia Syndromes With Cerebellar Hypoplasia

Inheritance	Gene locus	Gene	Clinical findings	OMIM
Autosomal recessive	2q31.1-q36.1	NEUROD1	Neonatal diabetes mellitus, NPCA, deafness, myopia	Rubio-Cabezas et al. [2010]
	8q11-12	CA8	NPCA, cognitive impairment	613227
	9p24	VLDLR	NPCA, moderate cognitive impairment, pes planus, strabismus, seizures, short stature	224050
	13q12.13	ATP8A2	NPCA, cognitive impairment	615268
	15q24-q26	ZNF592	NPCA, cognitive impairment, spasticity, speech disorder, severe microcephaly, short stature, optic atrophy, osmophilic skin vessels	606937
	17p	WDR81	NPCA, cognitive and speech impairment, coarse face, short stature, hirsutismus	610185
	19p13.3	Caytaxin	NPCA, marked cognitive impairment	601238
	9q34-qter	Unknown	NPCA, cognitive impairment, short stature, albinism	213200
	20q11-q13	Unknown	NPCA, mild spasticity, short stature	608029

NPCA, non-progressive cerebellar ataxia.

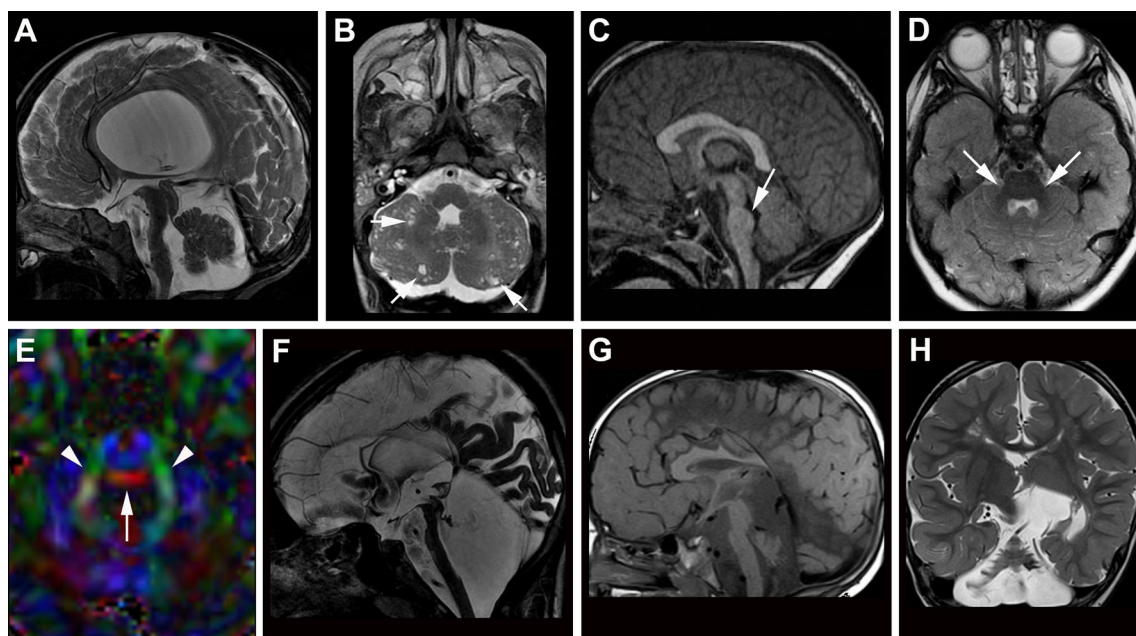


Figure 4. Pontocerebellar hypoplasia. **A:** Midsagittal T2-weighted image of a 5-month-old child with muscle-eye-brain disease (*POMGnT1* mutation) shows cerebellar hypoplasia, flattening of ventral pons, dysmorphic tectum and midbrain, abnormal concave posterior border of brainstem, enlarged fourth ventricle and supratentorial ventriculomegaly (reprinted with permission from Poretti et al., 2012d). **B:** Axial T2-weighted image of the same child demonstrates cerebellar hypoplasia and dysplasia with multiple cortical-subcortical cysts (arrows) in both cerebellar hemispheres and clefts in the dorsal and ventral pons (reprinted with permission from Poretti et al., 2012d). **C:** Midsagittal T1-weighted image of a 3-year-old girl with pontine tegmental cap dysplasia reveals a flat ventral pons and a cap covering dorsal pons (arrow) and protruding into the fourth ventricle. **D:** Axial T2-weighted image of the same child demonstrates mild cerebellar hypoplasia with hypoplastic middle cerebellar peduncles (arrows). **E:** Axial color-coded fractional anisotropy map (DTI) of the same patient shows an ectopic band of fibers in red (horizontal orientation) at the dorsal aspect of the pons (arrow) and small middle cerebellar peduncles in green (arrowheads anterior ↔ posterior orientation). **F:** Midsagittal T2-weighted image of a 15-year-old female with cerebellar agenesis demonstrates an empty enlarged posterior fossa with only rudimentary cerebellar remnants projecting posterior to the inferior colliculi and a marked hypoplasia of the pons (reprinted with permission from Poretti et al., 2009b). **G:** Midsagittal T1-weighted image of a 3-year-old boy born at 25 weeks of gestation shows a small posterior fossa, small vermis, pontine hypoplasia, and a thin corpus. **H:** Coronal T2-weighted image of the same child demonstrates marked reduction in size of the cerebellar hemispheres, which have a “skeletonized” appearance, a rather well preserved cerebellar vermis compared to the hemispheres and encephalomalacic changes in the supratentorial brain as sequelae of prematurity.

In lissencephaly, PCH is most typically found in association with *RELN*, *VLDLR*, and *TUBA1A* mutations [Jissendi-Tchofo et al., 2009b]. *RELN* (OMIM 257320) encodes an extracellular matrix-associated glycoprotein (reelin) that is secreted by Cajal–Retzius cells in the developing cerebral cortex. Reelin is critical for the regulation of neuronal migration during cortical and cerebellar development [Hong et al., 2000]. Affected children show severe developmental disabilities, microcephaly, seizures, and congenital lymphedema. Neuroimaging findings include pachygyria, severe CH with abnormal foliation and more severe involvement of the vermis compared to the hemispheric and pontine involvement.

VLDLR (OMIM 224050) encodes the very low-density lipoprotein receptor, which acts as a co-receptor for the *RELN* pathway. *VLDLR*-associated PCH is characterized by non-progressive cerebellar ataxia, moderate to profound intellectual disability, dysarthria, strabismus, and seizures [Boycott et al., 2005]. Neuroimaging shows simplified cerebral gyration, CH with reduced cerebellar foliation (milder compared to patients with *RELN* mutations) and pontine hypoplasia [Glass et al., 2005].

Genes involved in microtubule formation and function (*TUBA1A*, *TUBA8*, *TUBB2B*, *TUBB3*, and *TUBB5*) have been shown to cause malformations of cortical development including lissencephaly and polymicrogyria [Abdollahi et al., 2009; Poirier et al., 2010; Breuss et al., 2012; Cushion et al., 2013]. The clinical phenotype is wide ranging, from isolated congenital fibrosis of the extraocular muscles to severe intellectual disability, tetraspastic cerebral palsy, postnatal microcephaly, and early onset of therapy resistant seizures [Cushion et al., 2013]. Dysmorphic features are rare and other organs are not affected. The neuroimaging phenotype is also broad and overlaps between the different genetic causes (no phenotype-genotype correlation) [Cushion et al., 2013]. Cortical malformations include lissencephaly (usually

with an anterior to posterior gradient) and polymicrogyria (generalized, asymmetric with left-side preponderance, or only in the Perisylvian region). A dysmorphic appearance of the basal ganglia (mostly putamen and caudate) with absence of the anterior limb of the internal capsule is the most characteristic and consistent finding. Ventriculomegaly with abnormal shape of the frontal horns, as well as agenesis/dysgenesis of the corpus callosum and anterior commissure have also been described. Posterior fossa involvement includes different degrees of PCH, cerebellar and tectal dysplasia, and asymmetric midbrain and pons [Cushion et al., 2013]. The majority of mutations in tubulin genes are sporadic and de novo, but germline mosaicism and autosomal recessive inheritance have also been observed [Breuss et al., 2012].

Congenital disorders of glycosylation

PCH with superimposed CA is characteristic of congenital disorders of glycosylation type 1a (CDG1a; OMIM 212065) [Feraco et al., 2012]. Additional neuroimaging findings include T2-hyperintense signal of cerebellar cortex and subcortical white matter, transverse pontine fibers and median raphe, delayed myelination, atrophy of the supratentorial white matter and ventriculomegaly. CDG1a is a systemic disease and may present with hypotonia, developmental delay, abnormal fat distribution, coagulopathy, retinal degeneration, peripheral neuropathy, stroke-like episodes, or seizures [Freeze et al., 2014]. CDG1a is caused by mutations in *PMM2*, which plays a key role in N-glycosylation.

CASK-related PCH

CASK is on chromosome Xp11.4 (OMIM 300749) and encodes a multi-domain scaffolding protein that interacts with *TBR1* and regulates expression of genes involved in cortical development such as *RELN* [Najm et al., 2008]. *CASK* mutations occur de novo and more commonly affect females, presumably because they are lethal in males.

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Patients present with ataxia, nystagmus, postnatal microcephaly, intellectual disability, sensorineural hearing loss, and inconsistently cataract [Moog et al., 2011]. Neuroimaging findings include mild to severe PCH, reduced gyral pattern and unmyelinated corpus callosum in some patients [Takanashi et al., 2010].

Congenital muscular dystrophies due to defective α -dystroglycan glycosylation

“ α -dystroglycanopathies” are a group of congenital muscular dystrophies resulting from mutations in 15 genes responsible for the O- and rarely N-glycosylation of α -dystroglycan [Bonnemann et al., 2014].

Recessive mutations in these genes cause overlapping phenotypes characterized by muscle (weakness, hypotonia, and increased creatine kinase values), brain (intellectual disability, seizures, and tetraspasticity), and eye (microphthalmia, optic nerve hypoplasia, chorioretinal coloboma, cataract, glaucoma, or high myopia) involvement. Based on the severity of the findings, different phenotypes have been described in order of increasing severity: Fukuyama congenital muscular dystrophy, muscle-eye-brain disease, and Walker–Warburg syndrome.

Infratentorial neuroimaging findings include PCH, cerebellar dysplasia with cysts, dysplastic tectum, ventral pontine cleft, and ponto-mesencephalic kinking [Clement et al., 2008]. The cerebellar cortical and subcortical cysts are relatively specific and represent small areas of pia and subarachnoid space herniating inward through gaps in the pial limiting membrane. Supratentorial findings range from mild ventriculomegaly, diffuse periventricular white matter changes, and focal areas of polymicrogyria to severe hydrocephalus, generalized white matter signal changes and diffuse cortical abnormalities including cobblestone cortex [Clement et al., 2008]. Cerebellar dysplasia with cysts may be a form of dystroglycanopathy without cortical involvement [Poretti et al., 2014].

Pontine tegmental cap dysplasia

Pontine tegmental cap dysplasia (PTCD, OMIM 614688) is characterized by involvement of vestibulocochlear, facial, trigeminal and glossopharyngeal nerves with resultant hearing loss, trigeminal anesthesia, facial paralysis and difficulty in swallowing [Barth et al., 2007]. Systemic involvement with vertebral segmentation anomalies, rib malformations, and congenital heart defects has been observed. The prognosis appears to be highly variable, ranging between mild cognitive delay to severe disability [Barth et al., 2007]. PTCD is a sporadic malformation with no known genetic cause and no familial recurrence.

PTCD can be easily distinguished from other PCH's by the characteristic vaulted pontine tegmentum (the so-called "cap") and severe hypoplasia of the inferior and middle cerebellar peduncles [Jissendi-Tchofo et al., 2009a]. The degree of brainstem dysplasia seems to correlate with the developmental disability: mildly affected patients tend to have a rounded bump (the so-called cap) and those more severely affected tend to have a more angular brainstem kink (a so-called beak). Additional neuroimaging findings include duplicated internal auditory canals [Desai et al., 2011], hypoplastic cranial nerves, a "molar tooth like" aspect of the

ponto-mesencephalic junction, and absent inferior olivary prominence. DTI demonstrates absence of the transverse pontine fibers and presence of a dorsal transverse axonal band at the level of the "cap" in the dorsal pons [Jissendi-Tchofo et al., 2009a]. The dorsal ectopic axonal band and ectopic peripontine arcuate fibers most likely result from abnormal axonal guidance and/or neuronal migration [Caan et al., 2014].

Cerebellar agenesis

Cerebellar agenesis is characterized by near complete absence of cerebellar tissue and is the most severe form of CH.

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All patients are symptomatic and present with variable degrees of cerebellar dysfunction and cognitive impairment. Neonates should be evaluated for diabetes mellitus (*PTF1A* mutations).

Near-complete absence of cerebellar structures with remnants of the anterior vermis, floccules, and middle cerebellar peduncles are present on neuroimaging. In addition, pontine hypoplasia and a normal or enlarged posterior fossa may be seen.

Cerebellar disruption or injury secondary to prematurity

During the period from 28 gestational weeks to term, the surface area of the

cerebellar cortex increases more than 30-fold [Limperopoulos et al., 2005b]. This very rapid growth follows a well-programmed developmental process and is highly energy demanding. This places the developing cerebellum at high risk for injury, particularly during gestational weeks 24–32 [Messerschmidt et al., 2008]. Arrested cerebellar development with reduction in the final volume is the long-term sequela of cerebellar injuries in preterm infants.

Cerebellar injury occurs in up to 20% of preterm infants <32 weeks' gestational age [Volpe, 2009]. Impaired cerebellar development may be caused by primarily destructive injuries or secondary lesions. Primarily destructive injuries include hemorrhage and ischemia resulting in cerebellar tissue loss [Volpe, 2009]. Secondary lesions lead to cerebellar underdevelopment and can be direct (e.g., hemosiderin-blood products, glucocorticoid exposure, and under nutrition) or indirect (impaired trans-synaptic trophic effects) [Volpe, 2009]. General perinatal risk factors include prolonged ventilation, pressor support, post-hemorrhagic hydrocephalus, and infratentorial hemosiderin deposition [Limperopoulos et al., 2005a].

The typical neuroimaging findings are a small cerebellum with predominant involvement of "skeletonized" cerebellar hemispheres, flattened ventral aspect of pons and an enlarged balloon-shaped fourth ventricle [Messerschmidt et al., 2005]. Hemosiderin deposition may be present and is best demonstrated by SWI. Surviving preterm infants have a high prevalence of motor, cognitive, and behavioral disabilities including autistic features [Volpe, 2009]. Attributing PCH to prematurity should be approached with caution, given the 25% recurrence risk for some genetic causes of PCH. History and additional neuroimaging findings such as hydrocephalus and sequelae of intraventricular hemorrhages or periventricular venous infarction may be helpful to differentiate between genetic PCH from PCH secondary to prematurity.

DIAGNOSTIC WORK-UP OF CEREBELLAR HYPOPLASIA

CH is a non-specific finding that can result from several etiologies. After a complete history and physical examination, the most important evaluation is a comprehensive review of the brain MRI by a physician experienced in the diagnosis of hindbrain malformations. The type of CH, other neuroimaging findings and clinical findings may suggest a specific disease or help to plan further investigations and interpret their results. Serum creatine kinase is an inexpensive test to evaluate for α -dystroglycanopathies. Similarly, isoelectric focusing of serum transferrin should be performed if CDG is suspected, and a 7-dehydrocholesterol level to evaluate for SLO. Electroencephalogram is not helpful unless seizures are suspected. Chromosome array and whole-exome sequencing may be indicated if dysmorphic signs or other features suggest a genetic disorder. One study has reported a high yield of exome sequencing in pediatric patients with ataxia [Sawyer et al., 2014].

CONCLUSIONS

Recent advances in neuroimaging and genetic testing have greatly improved the classification of CH disorders and identification of their causes. Many patients and their families now have access to diagnostic, carrier and prenatal testing. In addition, information on outcomes in patients with well-defined hindbrain malformations is continually improving. Identifying the genetic causes of these disorders is a major step toward understanding the underlying mechanisms that will be the best targets for specific therapies in the future.

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